VINAY MUPPALLA

mvinay988579@gmail.com | +1 201-355-7269 | linkedin.com/in/muppalla-vinay

Portfolio | Research evidence | github.com/vinaychowdary9946-blip

RESEARCH INTERESTS

Reliable data systems; empirical software engineering; workflow recovery; trustworthy agent infrastructure; database integrity and fault-injection testing.

EDUCATION

M.S., Computer Information Systems - Rivier University | GPA: 3.79/4.0

B.Tech, Computer Science & Engineering - KITS, Guntur | CGPA: 7.2/10

INDEPENDENT SYSTEMS STUDY

Effect Boundary Lab | 2026: Investigates retries after a remote service forgets its duplicate-protection receipt. Python, HTTP, separate SQLite stores and subprocess fault injection; raw evidence and a reproducible protocol are publicly available.

Published artifact reports 1,152 controlled trials. Guarded retry had zero excess effects and zero false confirmations across 288 trials, while quarantining 72 operations. These deterministic synthetic results are scoped to the experimental fixture.

RESEARCH QUESTION AND METHOD

When is a restarted worker justified in repeating an external operation after the destination receipt expires? The study compares four recovery policies across eight scripted scenarios and three synthetic payload variants, using a loopback HTTP destination, separate SQLite stores, real process exits, raw trial logs and an independent verifier.

A paired witness records identical durable client intent with zero versus one remote effect. It illustrates why unresolved outcomes require additional destination evidence before a safe automatic decision can be made.

SCOPE AND NEXT RESEARCH

The study is an independent technical artifact, not a peer-reviewed publication. It uses virtual time, synthetic append effects and sequential clients. Existing idempotency, receipts and quarantine methods are not claimed as new algorithms.

Proposed extensions: authenticated destination reconciliation; concurrent clients and longer fault schedules; independent replication; evaluation against multiple destination contracts.

TECHNICAL SKILLS

Python, Java, SQL, SQLite, Git; object-oriented programming, data structures, algorithms, HTTP APIs, unit and integration testing, debugging, SDLC and documentation. Additional tools: Excel, SAP GUI and ERP systems.

SELECTED SOFTWARE PROJECTS

Resilient Job Queue: Durable background processing with leases and retries; a documented process-crash lab examines duplicate side effects and recovery. Python backend, local API and interactive browser replay.

Dependency Planner: Graph-based task scheduling with cycle detection, critical-path analysis and slack calculations; interactive demonstration and Python implementation.

Inventory Ledger: Event-history-based stock reconstruction with duplicate-command handling, overselling checks and version-conflict detection.

ACADEMIC PROJECTS

SMART START - IoT Security System | Team Lead: Authentication-enabled vehicle ignition, sensor integration, safety mechanisms, debugging and system validation.

Business Acquisition Case Study: Supply-chain and operations analysis, workflow improvements and documented recommendations.

RESEARCH MATERIALS

Technical report, discussion brief, protocol, related work, raw results and reproducible code bundle: https://vinaychowdary9946-blip.github.io/research/